

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A Analytical

Problem Solving

1. A coin is tossed 10 times and the outcomes are: H

H T H H H T H H T

Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

Aim:

To estimate the probability of getting Heads (p) using the Maximum Likelihood Estimation (MLE) method.

Given Data

Coin toss outcomes:

H H T H H H T H H T

Total number of tosses (n) = 10

Number of Heads (H) = 7

Number of Tails (T) = 3

Formula

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Number of Tosses}}$$

Calculation

$$\hat{p} = \frac{7}{10} = 0.7$$

Therefore,

Estimated probability of Heads (p) = 0.7

Estimated probability of Tails (1 - p) = 0.3

Analysis

Out of 10 coin tosses, Heads occurred 7 times.

MLE estimates the probability by taking the observed frequency of Heads.

Hence, the estimated probability of getting Heads is 0.7 (70%).

Result

The Maximum Likelihood Estimate (MLE) of the probability of getting Heads is:

p=0.7

Coin Tosses (10)

H H T H H H T H H T
| | | | | | | |
✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓

Heads = 7
Tails = 3

MLE:
 $p = \text{Heads} / \text{Total Tosses}$
 $= 7 / 10$
 $= 0.7$

2. Out of 50 students, 42 passed an exam.

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Aim:

To estimate the probability that a student passes an exam using Maximum Likelihood Estimation (MLE) assuming a Bernoulli distribution.

Given Data

Total number of students (n) = 50

Number of students who passed = 42

Number of students who failed = 8

Formula

For a Bernoulli distribution,

$$\hat{p} = \frac{\text{Number of Passes}}{\text{Total Number of Students}}$$

Calculation

$$\hat{p} = \frac{42}{50} = 0.84$$

Therefore,

Estimated probability of passing (p) = 0.84

Estimated probability of failing (1 - p) = 0.16

Analysis

Out of 50 students, 42 students passed the exam.

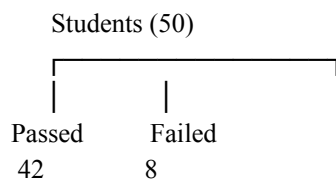
MLE estimates the probability by using the observed proportion of passes.

Hence, the estimated probability that a student passes is 0.84 (84%).

Result

The Maximum Likelihood Estimate (MLE) of the probability that a student passes the exam is:

p=0.84



MLE:

$p = \text{Passes} / \text{Total Students}$
 $= 42 / 50$
 $= 0.84$

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.

Aim:

To estimate the probability that a light bulb is defective using Maximum Likelihood Estimation (MLE).

Given Data

Total number of bulbs (n) = 100

Number of defective bulbs = 8

Number of non-defective bulbs = 92

Formula

$$\hat{p} = \frac{\text{Number of Defective Bulbs}}{\text{Total Number of Bulbs}}$$

Calculation

$$\hat{p} = \frac{8}{100} = 0.08$$

Therefore,

Estimated probability of a defective bulb (p) = 0.08

Estimated probability of a non-defective bulb ($1 - p$) = 0.92

Analysis

Out of 100 bulbs, 8 bulbs are defective.

MLE estimates the probability using the observed proportion of defective bulbs.

Hence, the estimated probability that a bulb is defective is 0.08 (8%).

Result

The Maximum Likelihood Estimate (MLE) of the probability that a bulb is defective is: $p=0.08$

Light Bulbs (100)

┌───────────┐	
│ │	
Defective	Non-Defective
8	92

MLE:

$$p = \text{Defective} / \text{Total Bulbs}$$

$$= 8 / 100$$

$$= 0.08$$

4. A biased die is rolled 60 times. The number 6 appears 18 times.

Estimate the probability of rolling a 6 using MLE.

Aim:

To estimate the probability of rolling a 6 using Maximum Likelihood Estimation (MLE).

Given Data

Total number of die rolls (n) = 60

Number of times 6 appeared = 18

Number of times other numbers (1–5) appeared = 42

Formula

$$\hat{p} = \frac{\text{Number of times 6 appeared}}{\text{Total number of rolls}}$$

Calculation

$$\hat{p} = \frac{18}{60} = 0.30$$

Therefore,

Estimated probability of rolling a 6 (p) = 0.30

Estimated probability of not rolling a 6 ($1 - p$) = 0.70

Analysis

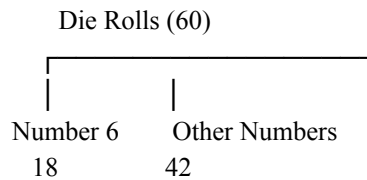
Out of 60 die rolls, the number 6 appeared 18 times.

MLE estimates the probability using the observed proportion of rolling a 6.

Hence, the estimated probability of rolling a 6 is 0.30 (30%).

Result

The Maximum Likelihood Estimate (MLE) of the probability of rolling a 6 is: $p=0.30$



MLE:

$$\begin{aligned} p &= \text{Number of 6's} / \text{Total Rolls} \\ &= 18 / 60 \\ &= 0.30 \end{aligned}$$

5. Given:

Input (x) = 4

Weight (w) = 2

Actual Output = 10

Prediction:

$$y = w \times x$$

1. Find the predicted output.
2. Calculate the error.
3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

I M.Ajitha(192525391)____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation